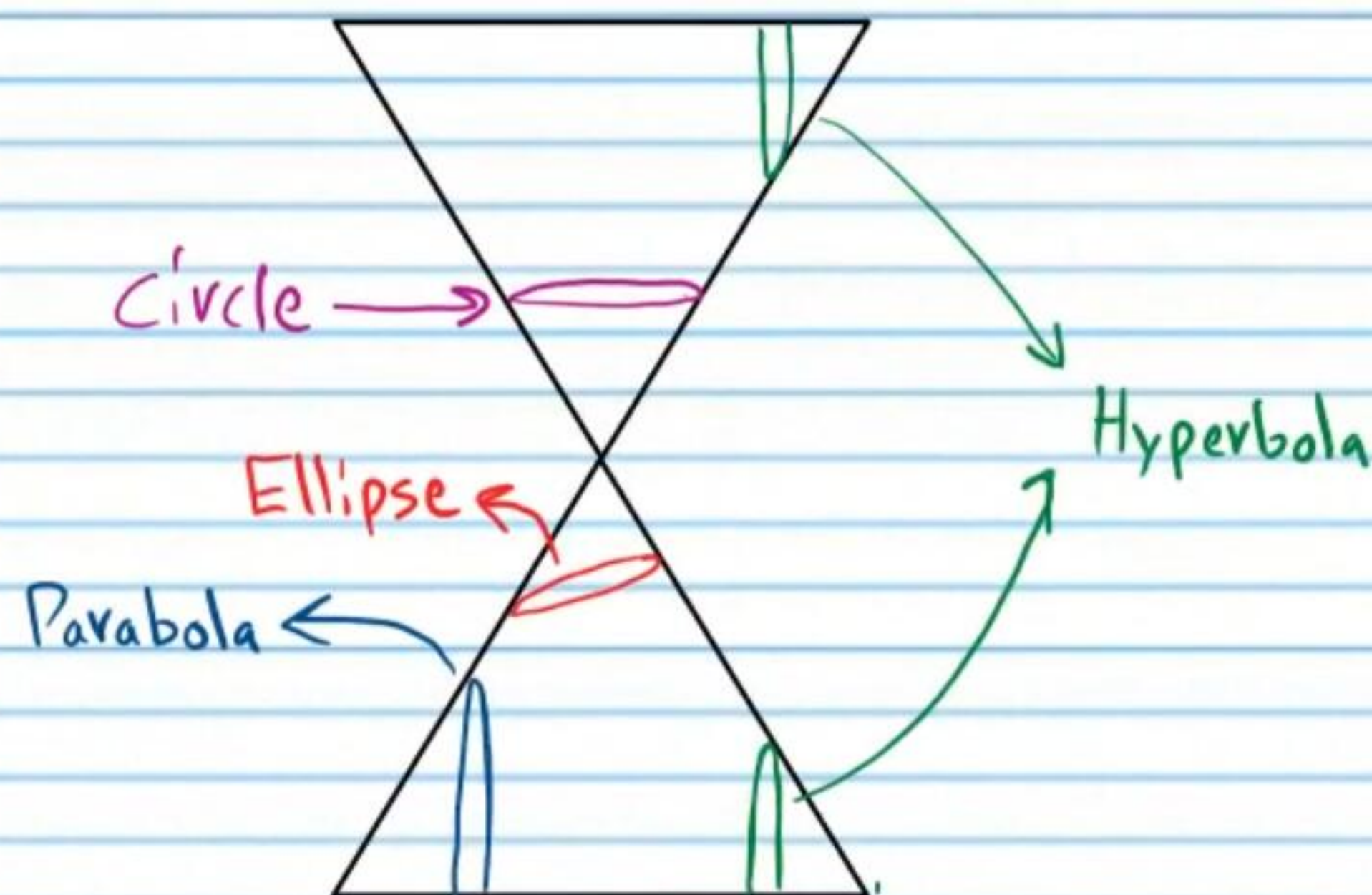


The Conics (Part III)



Equations of the Conics

Parabola

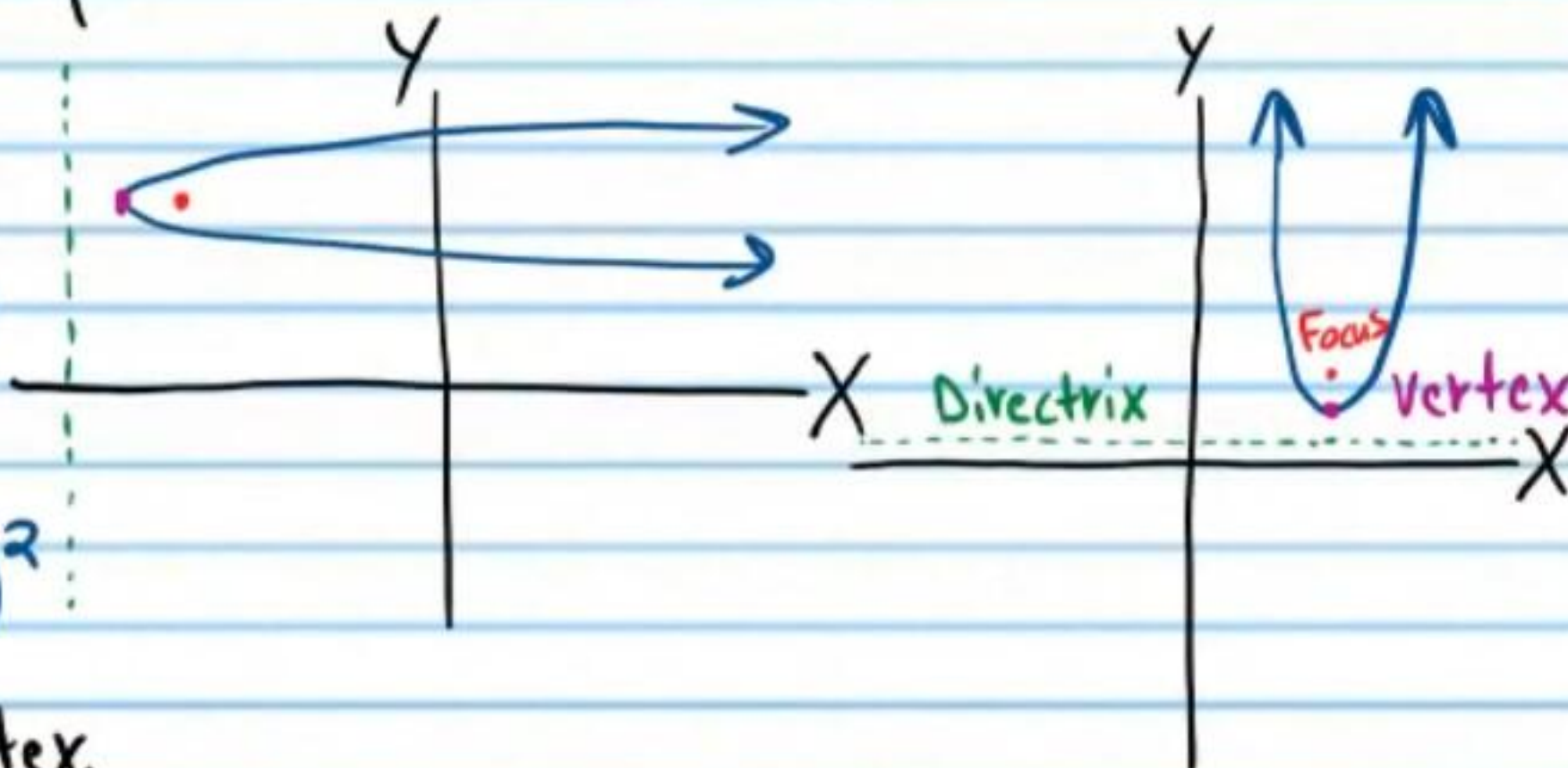
Vertical
 $y - k = a(x - h)^2$

Horizontal
 $x - h = a(y - k)^2$

(h, k) is the vertex.

$$a = \frac{1}{4p}$$

p is the distance from the vertex to the focus or directrix.



Other Forms

Standard

$$y = ax^2 + bx + c$$

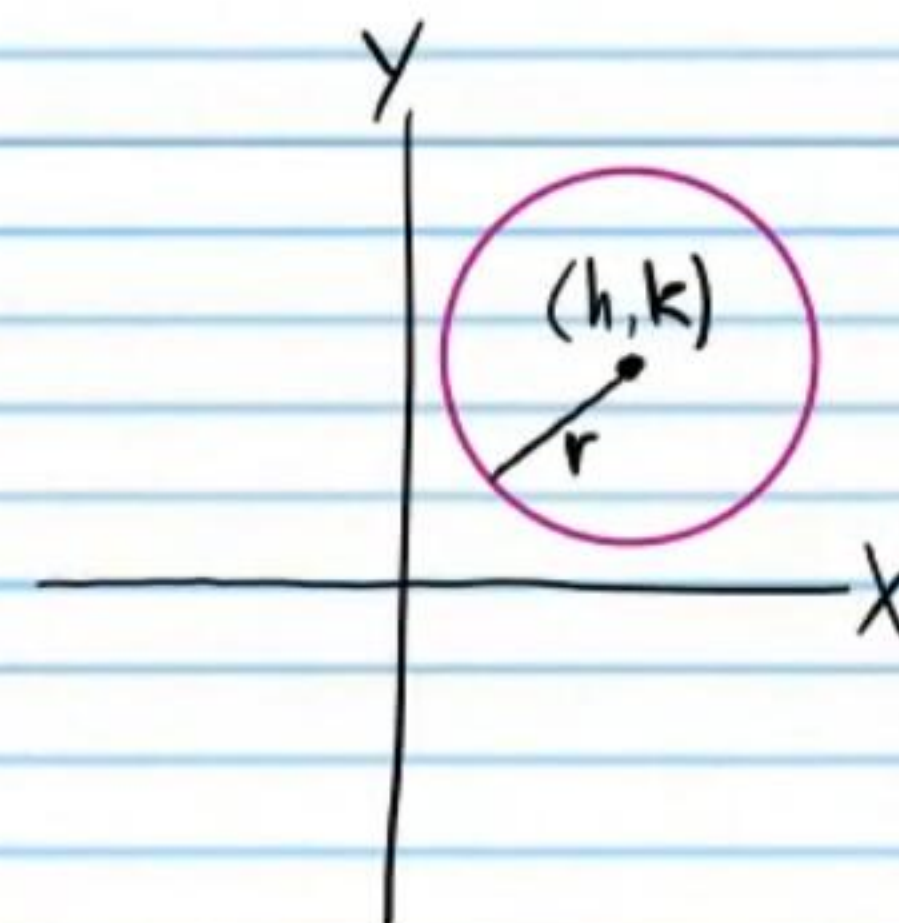
Intercept

$$y = a(x - m)(x - n)$$

Circle

$$(x - h)^2 + (y - k)^2 = r^2$$

(h, k) is the center.
 r is the radius.



Ellipse

Horizontal

$$\frac{(x - h)^2}{a^2} + \frac{(y - k)^2}{b^2} = 1$$

Vertical

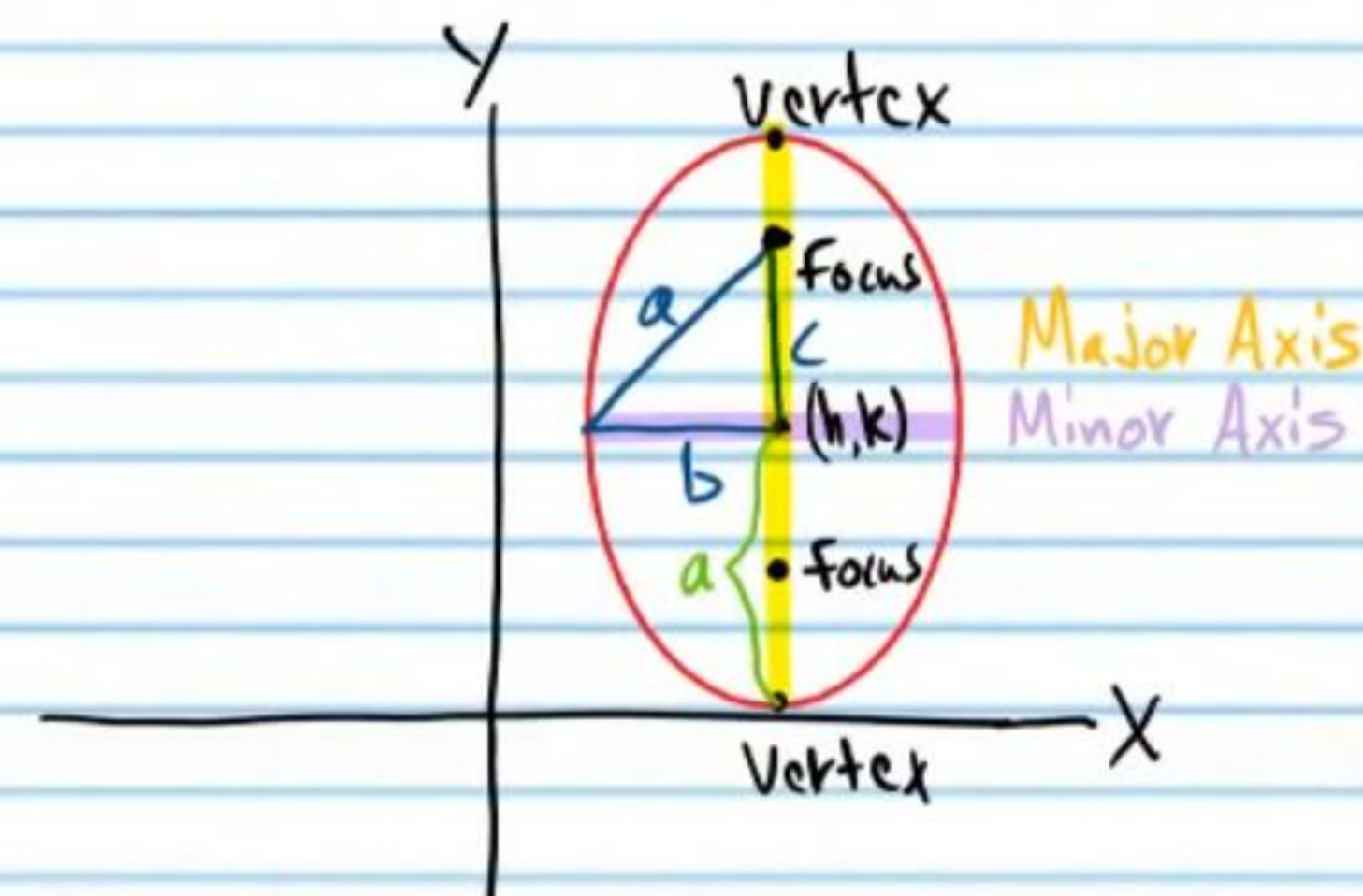
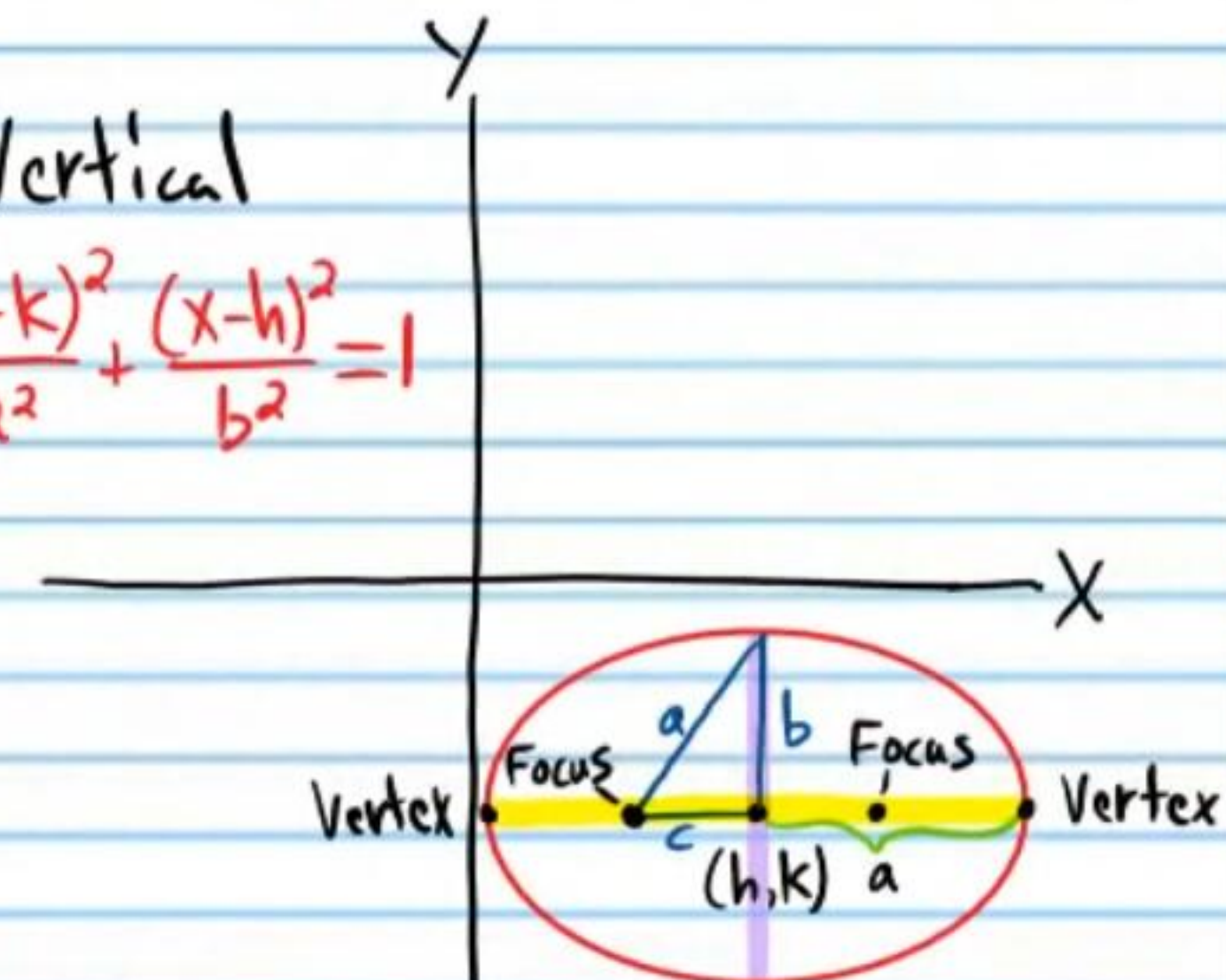
$$\frac{(y - k)^2}{a^2} + \frac{(x - h)^2}{b^2} = 1$$

Vertical

(h, k) is the center.

$$c^2 + b^2 = a^2$$

c is the distance along the major axis from the center to the foci.
 a is the distance from the center to the vertices. $2a$ is the length of the major axis.
 $2b$ is the length of the minor axis.



Hyperbola

Opens Right and Left

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$$

Opens Up and Down

$$\frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1$$

(h, k) is the center.

$$a^2 + b^2 = c^2$$

c is the distance from the center to the foci. a is the distance from the center to the vertices.

Asymptotes Opening Up/Down

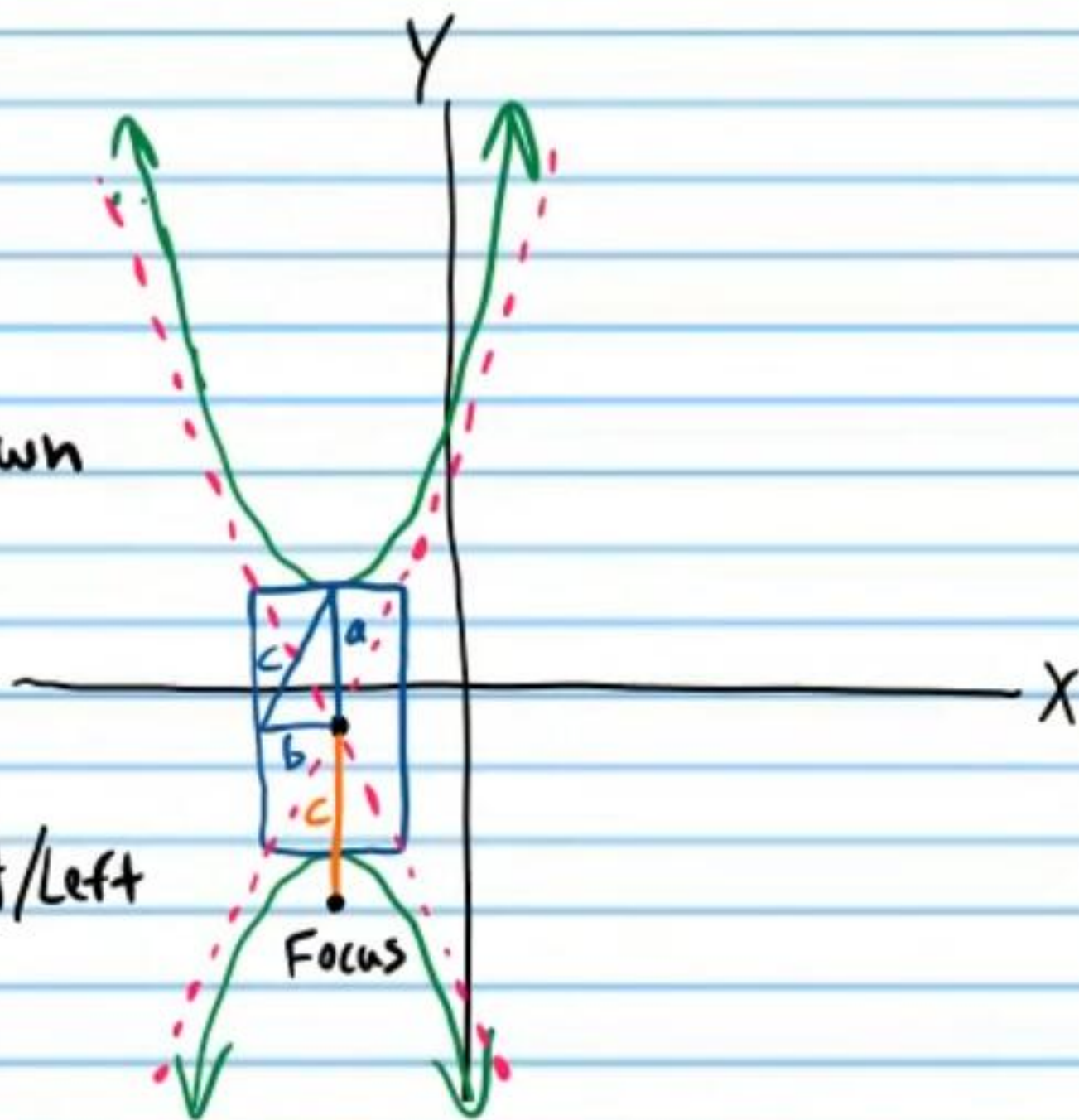
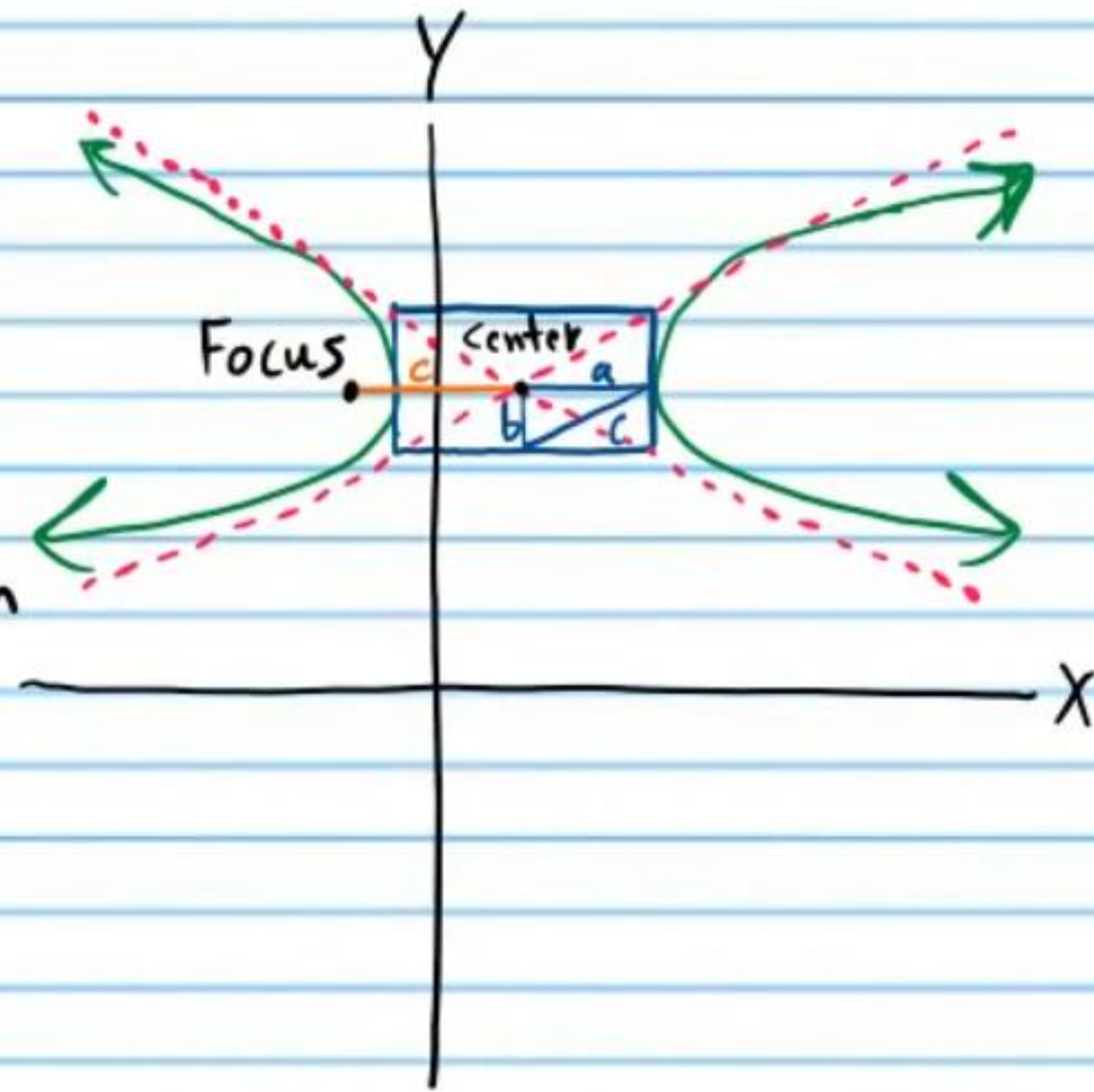
$$y - k = \frac{a}{b}(x - h)$$

$$y - k = -\frac{a}{b}(x - h)$$

Asymptotes Opening Right/Left

$$x - h = \frac{b}{a}(y - k)$$

$$x - h = -\frac{b}{a}(y - k)$$



Changing the Form of Equations in Order to Graph Them

Graph the following equations. *You must first convert to a recognizable form.*

① $x^2 + y^2 - 2x + 6y - 15 = 0$

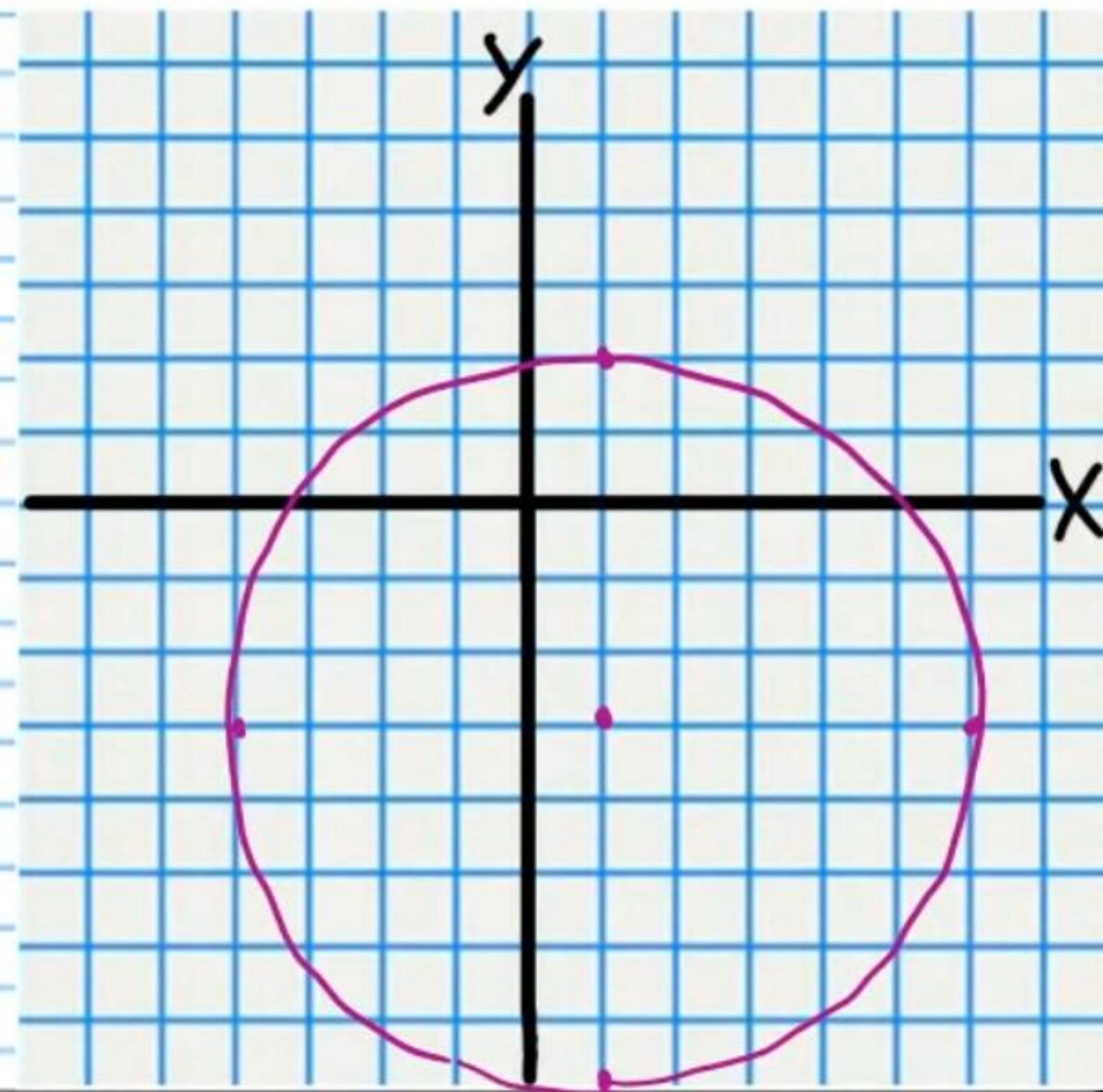
$$x^2 - 2x + y^2 + 6y = 15$$

$$x^2 - 2x + \left(-\frac{2}{2}\right)^2 + y^2 + 6y + \left(\frac{6}{2}\right)^2 = 15 + \left(-\frac{2}{2}\right)^2 + \left(\frac{6}{2}\right)^2$$

$$x^2 - 2x + 1 + y^2 + 6y + 9 = 15 + 1 + 9$$

$$(x-1)(x-1) + (y+3)(y+3) = 25$$

$$(x-1)^2 + (y+3)^2 = 25$$



$$\textcircled{2} 16x^2 + y^2 + 64x + 8y + 64 = 0$$

$$16x^2 + 64x + y^2 + 8y = -64$$

$$16(x^2 + 4x) + y^2 + 8y = -64$$

$$16(x^2 + 4x + (\frac{4}{2})^2) + y^2 + 8y + (\frac{8}{2})^2 = -64 + (\frac{8}{2})^2 + 16(\frac{4}{2})^2$$

$$16(x^2 + 4x + 4) + y^2 + 8y + 16 = -64 + 16 + 16(4)$$

$$16(x+2)(x+2) + (y+4)(y+4) = 16$$

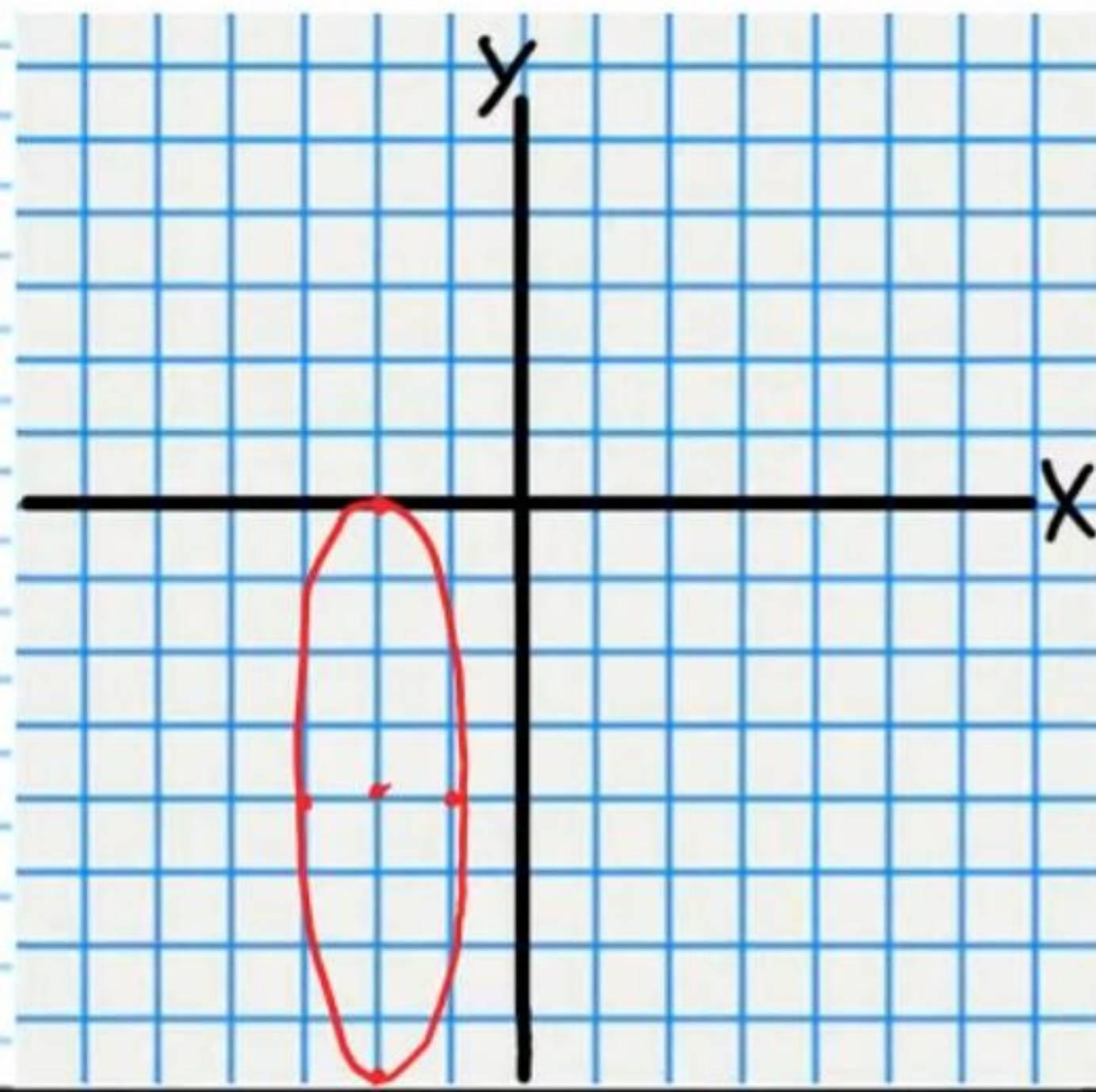
$$16(x+2)^2 + (y+4)^2 = 16$$

$$\frac{16(x+2)^2}{16} + \frac{(y+4)^2}{16} = \frac{16}{16}$$

$$\frac{16(x+2)^2}{16} + \frac{(y+4)^2}{16} = 1$$

$$(x+2)^2 + \frac{(y+4)^2}{16} = 1$$

$$\frac{(x+2)^2}{1^2} + \frac{(y+4)^2}{4^2} = 1$$



$$\textcircled{3} 4y^2 - 9x^2 - 24y + 36x - 36 = 0$$

$$4y^2 - 24y - 9x^2 + 36x = 36$$

$$4(y^2 - 6y) - 9(x^2 - 4x) = 36$$

$$4(y^2 - 6y + (\frac{-6}{2})^2) - 9(x^2 - 4x + (\frac{-4}{2})^2) = 36 + 4(\frac{-6}{2})^2 - 9(\frac{-4}{2})^2$$

$$4(y^2 - 6y + 9) - 9(x^2 - 4x + 4) = 36 + 36 - 36$$

$$4(y-3)(y-3) - 9(x-2)(x-2) = 36$$

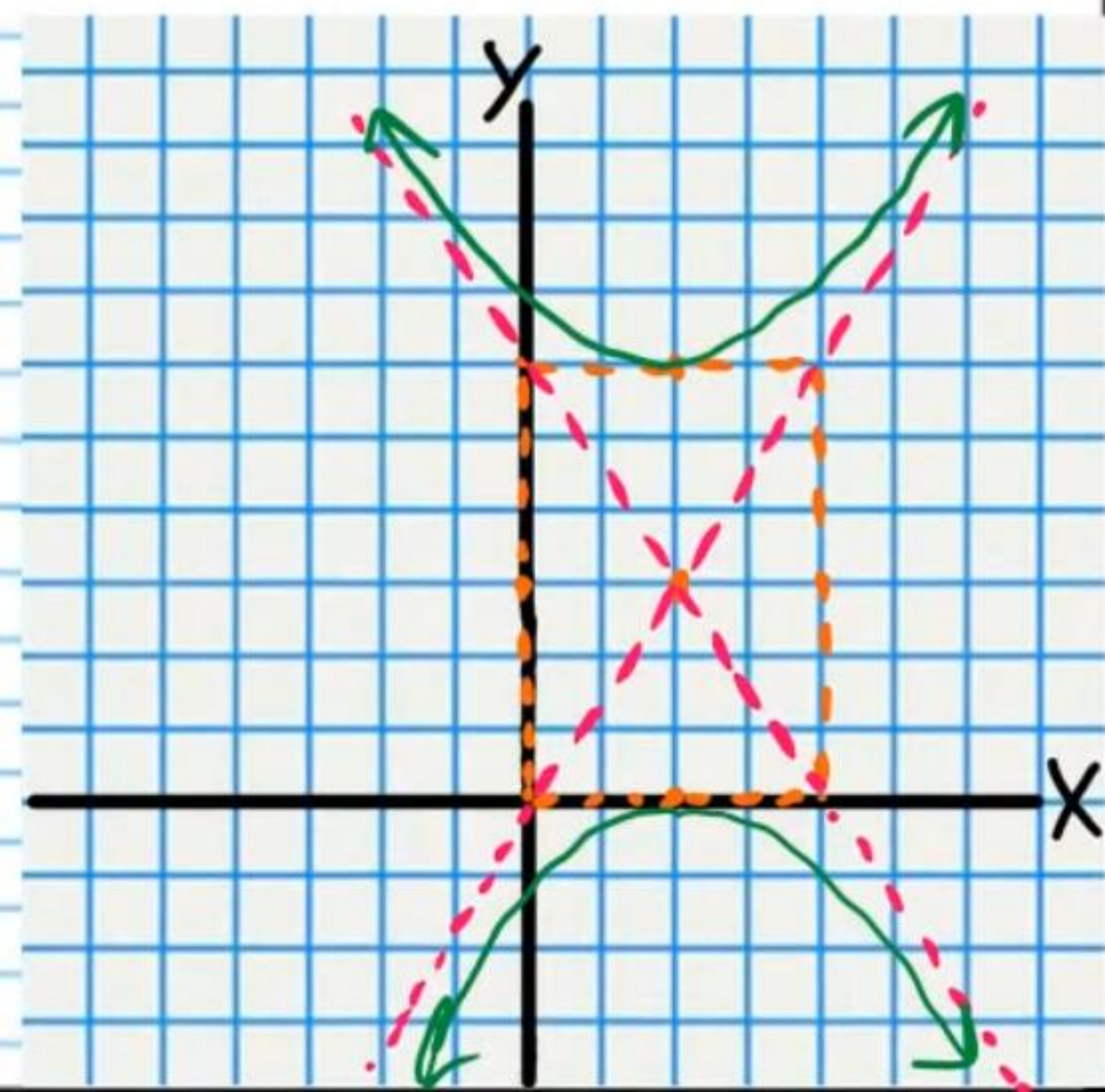
$$4(y-3)^2 - 9(x-2)^2 = 36$$

$$\frac{4(y-3)^2}{36} - \frac{9(x-2)^2}{36} = \frac{36}{36}$$

$$\frac{4(y-3)^2}{36} - \frac{9(x-2)^2}{36} = 1$$

$$\frac{(y-3)^2}{9} - \frac{(x-2)^2}{4} = 1$$

$$\frac{(y-3)^2}{3^2} - \frac{(x-2)^2}{2^2} = 1$$



$$\textcircled{4} y = 2x^2 + 20x + 52$$

$$y - 52 = 2x^2 + 20x$$

$$y - 52 = 2(x^2 + 10x)$$

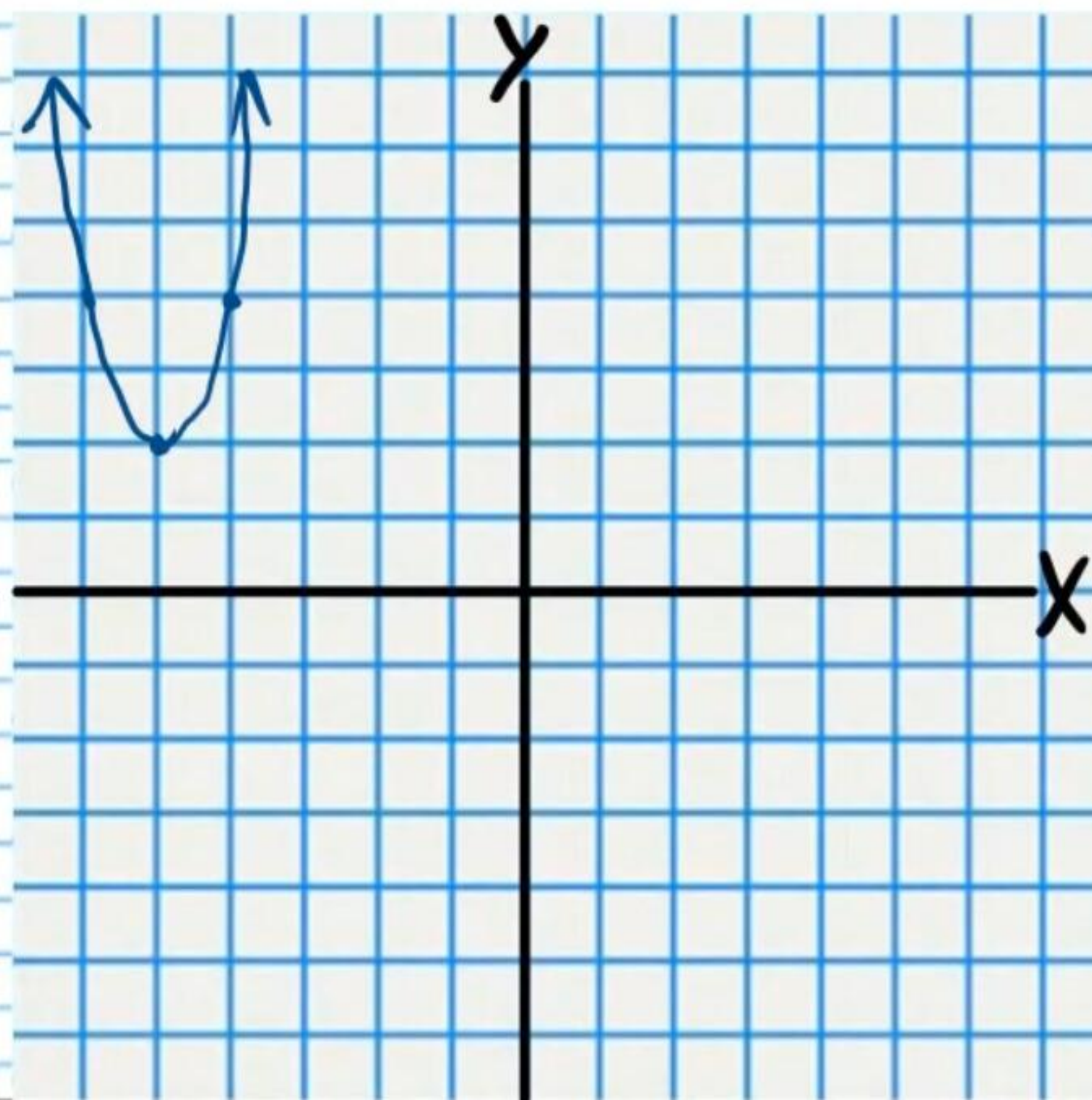
$$y - 52 + 2\left(\frac{10}{2}\right)^2 = 2\left(x^2 + 10x + \left(\frac{10}{2}\right)^2\right)$$

$$y - 52 + 2(25) = 2(x^2 + 10x + 25)$$

$$y - 2 = 2(x + 5)(x + 5)$$

$$y - 2 = 2(x + 5)^2$$

$$V(-5, 2)$$



$$y = ax^2 + bx + c$$

$$y = a(x - h)^2 + k$$

$$y = a(x - m)(x - n)$$

$$\bullet \textcircled{5} 25x^2 + 36y^2 + 50x + 72y - 839 = 0$$

$$25x^2 + 50x + 36y^2 + 72y = 839$$

$$25(x^2 + 2x) + 36(y^2 + 2y) = 839$$

$$25\left(x^2 + 2x + \left(\frac{2}{2}\right)^2\right) + 36\left(y^2 + 2y + \left(\frac{2}{2}\right)^2\right) = 839 + 25\left(\frac{2}{2}\right)^2 + 36\left(\frac{2}{2}\right)^2$$

$$25(x^2 + 2x + 1) + 36(y^2 + 2y + 1) = 839 + 25(1) + 36(1)$$

$$25(x + 1)(x + 1) + 36(y + 1)(y + 1) = 839 + 25 + 36$$

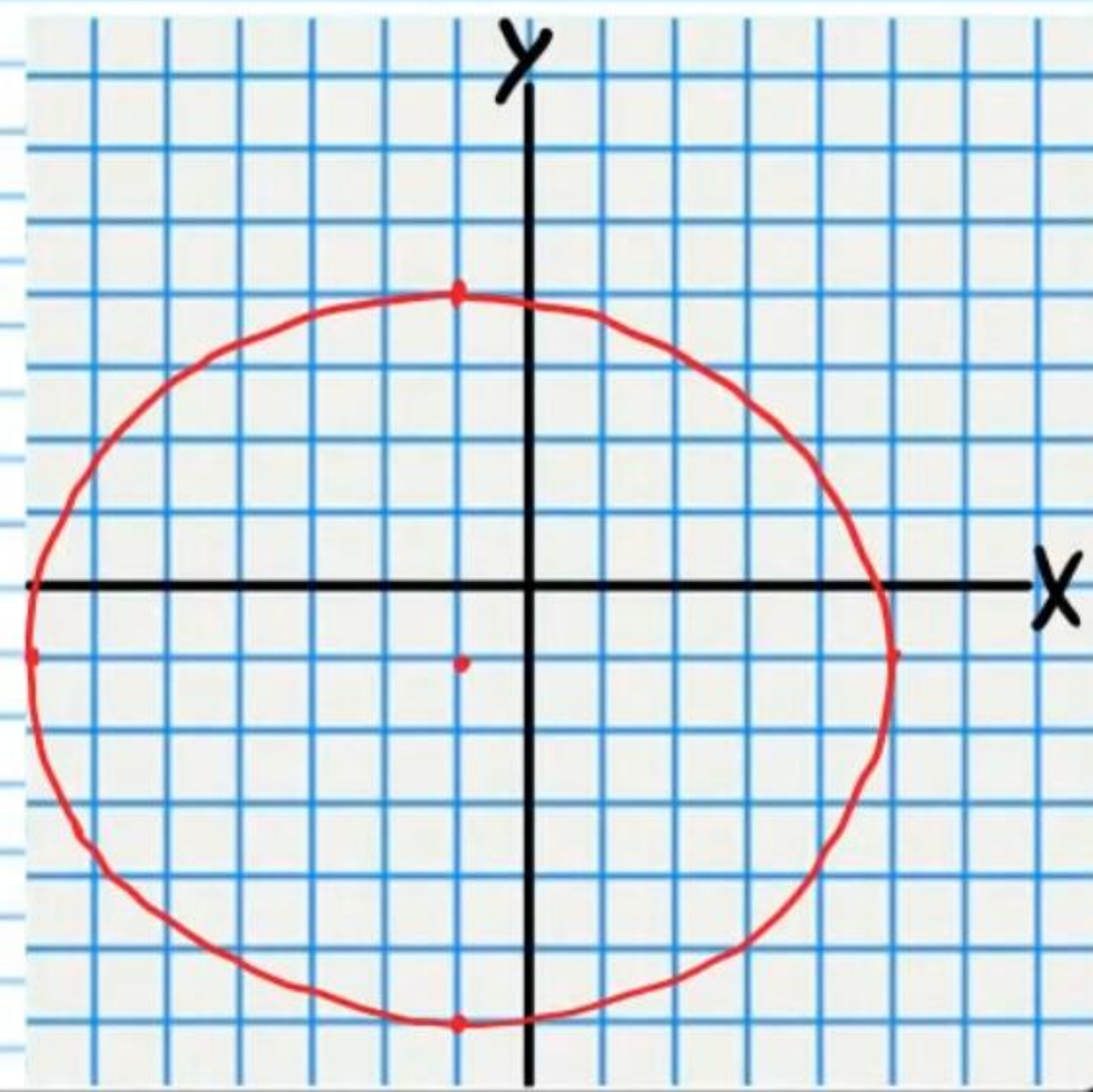
$$25(x + 1)^2 + 36(y + 1)^2 = 900$$

$$\frac{25(x + 1)^2}{900} + \frac{36(y + 1)^2}{900} = \frac{900}{900}$$

$$\frac{25(x + 1)^2}{900} + \frac{36(y + 1)^2}{900} = 1$$

$$\frac{(x + 1)^2}{36} + \frac{(y + 1)^2}{25} = 1$$

$$\frac{(x + 1)^2}{6^2} + \frac{(y + 1)^2}{5^2} = 1$$



$$\textcircled{6} x^2 - 9y^2 + 8x + 72y - 137 = 0$$

$$x^2 + 8x - 9y^2 + 72y = 137$$

$$x^2 + 8x - 9(y^2 - 8y) = 137$$

$$x^2 + 8x + \left(\frac{8}{2}\right)^2 - 9\left(y^2 - 8y + \left(\frac{-8}{2}\right)^2\right) = 137 + \left(\frac{8}{2}\right)^2 - 9\left(\frac{-8}{2}\right)^2$$

$$x^2 + 8x + 16 - 9(y^2 - 8y + 16) = 137 + 16 - 9(16)$$

$$(x+4)(x+4) - 9(y-4)(y-4) = 137 + 16 - 144$$

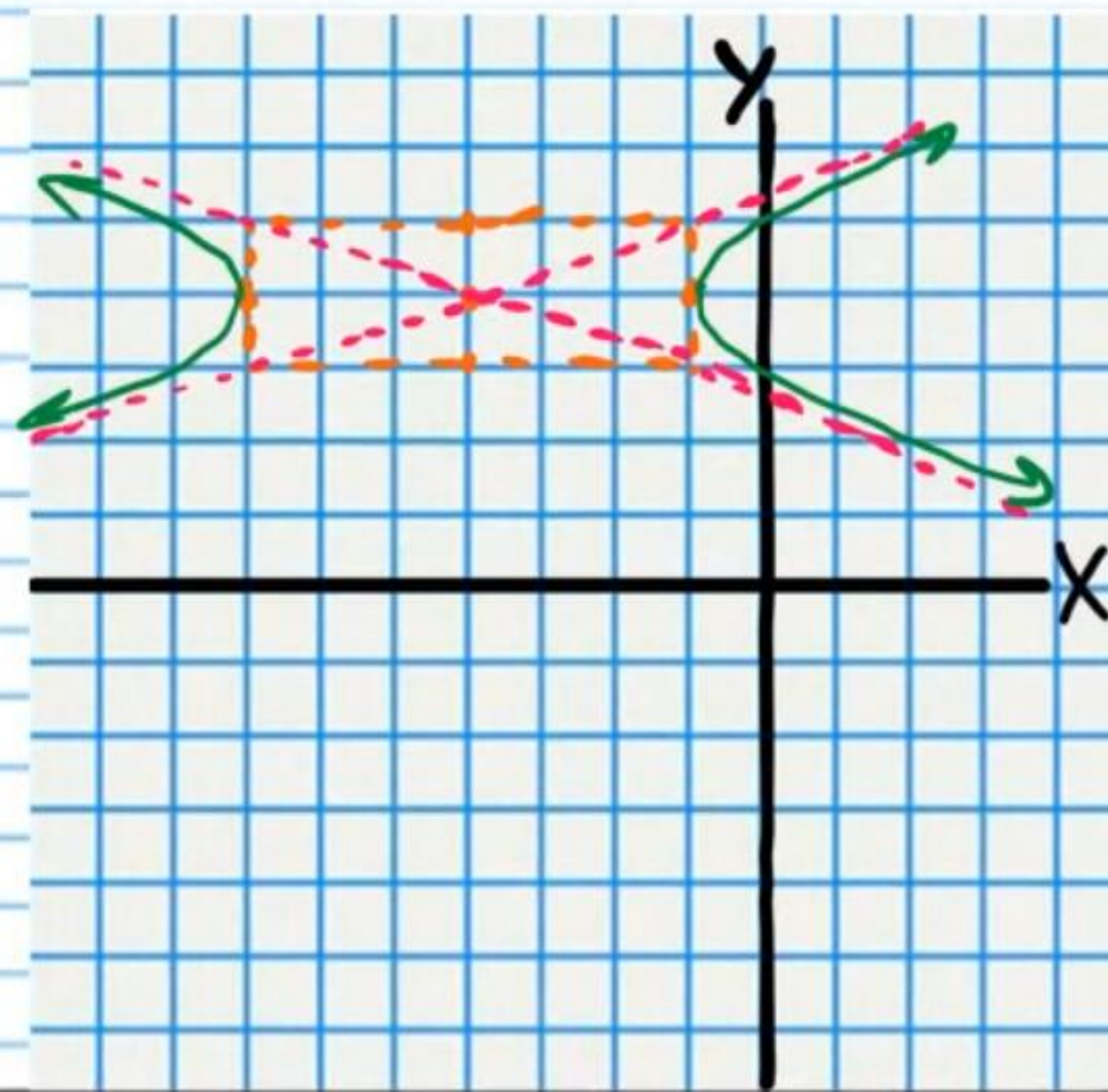
$$(x+4)^2 - 9(y-4)^2 = 9$$

$$\frac{(x+4)^2 - 9(y-4)^2}{9} = \frac{9}{9}$$

$$\frac{(x+4)^2}{9} - \frac{9(y-4)^2}{9} = 1$$

$$\frac{(x+4)^2}{9} - (y-4)^2 = 1$$

$$\frac{(x+4)^2}{3^2} - \frac{(y-4)^2}{1^2} = 1$$



* Start by multiplying both sides of the equation by -2 .

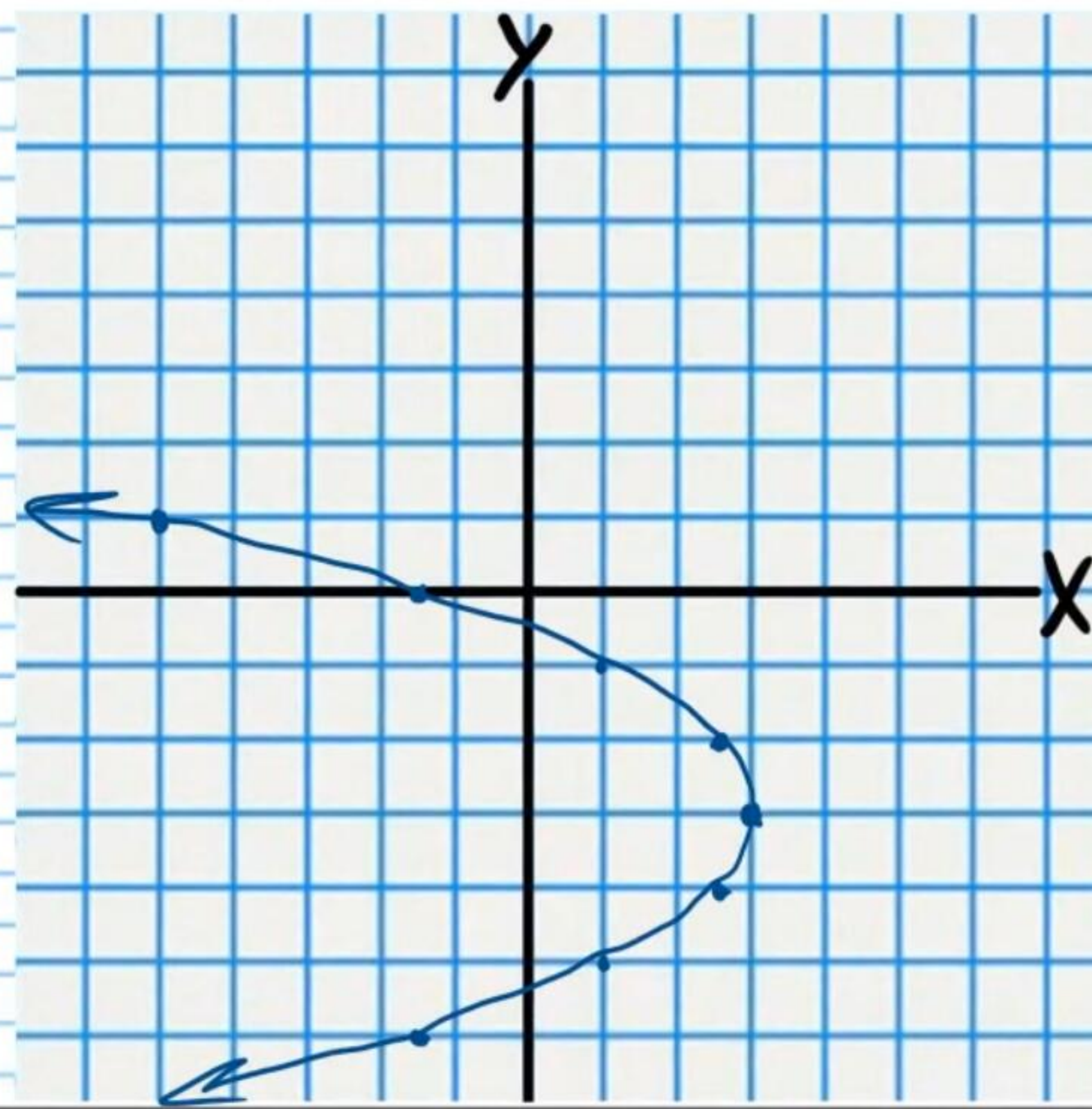
$$\textcircled{7} x - 3 = -\frac{1}{2}y^2 - 3y - \frac{9}{2}$$

$$-2(x-3) = y^2 + 6y + 9$$

$$-2(x-3) = (y+3)(y+3)$$

$$-2(x-3) = (y+3)^2$$

$$x - 3 = -\frac{1}{2}(y+3)^2$$



$$\textcircled{8} \ x^2 + y^2 - 4x - 4y - 8 = 0$$

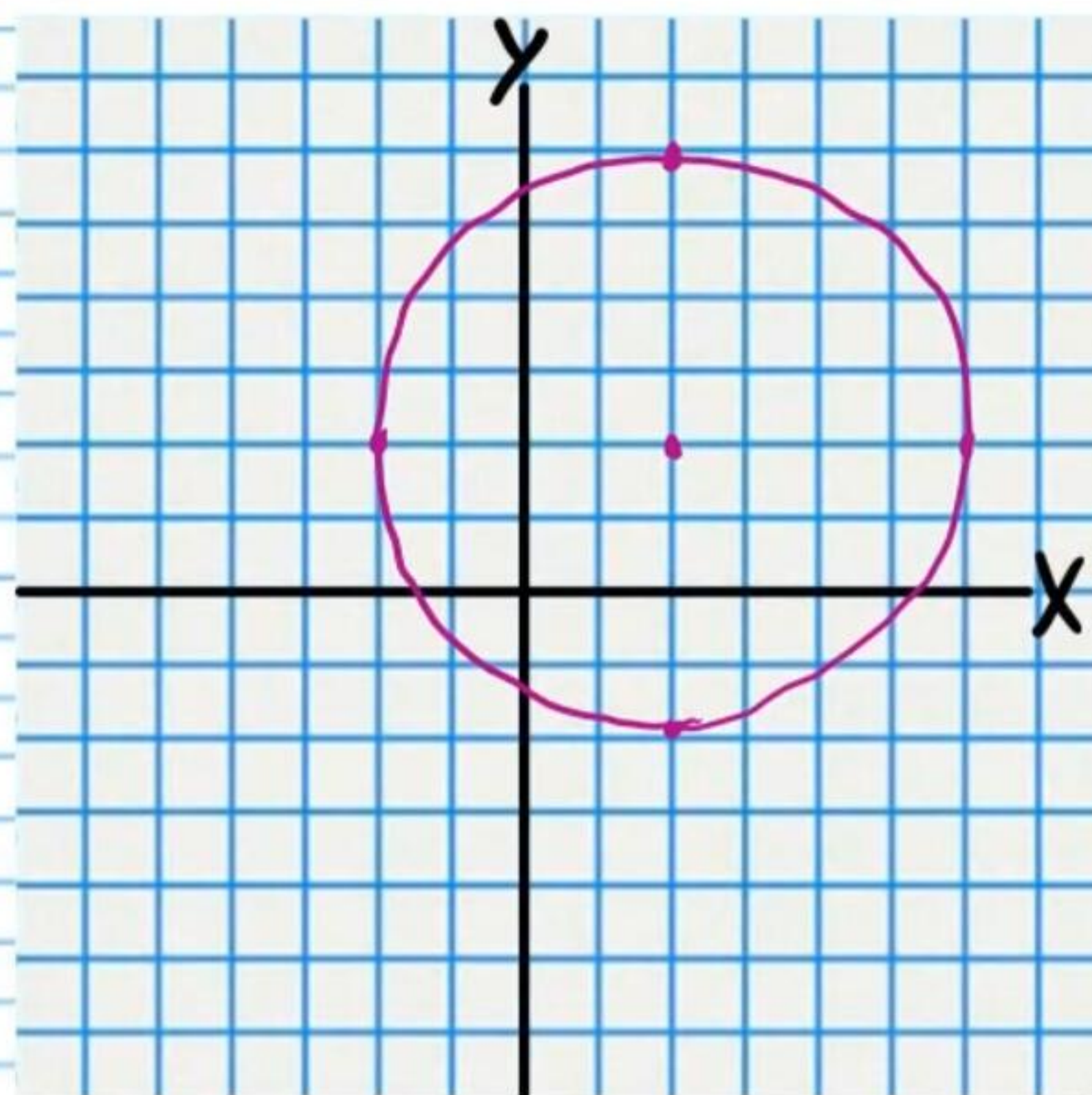
$$x^2 - 4x + y^2 - 4y = 8$$

$$x^2 - 4x + \left(-\frac{4}{2}\right)^2 + y^2 - 4y + \left(-\frac{4}{2}\right)^2 = 8 + \left(-\frac{4}{2}\right)^2 + \left(-\frac{4}{2}\right)^2$$

$$x^2 - 4x + 4 + y^2 - 4y + 4 = 8 + 4 + 4$$

$$(x-2)(x-2) + (y-2)(y-2) = 16$$

$$(x-2)^2 + (y-2)^2 = 16$$



$$\bullet \textcircled{9} \ 10x^2 - 10y^2 + 20x = 0$$

$$x^2 - y^2 + 2x = 0$$

$$x^2 + 2x - y^2 = 0$$

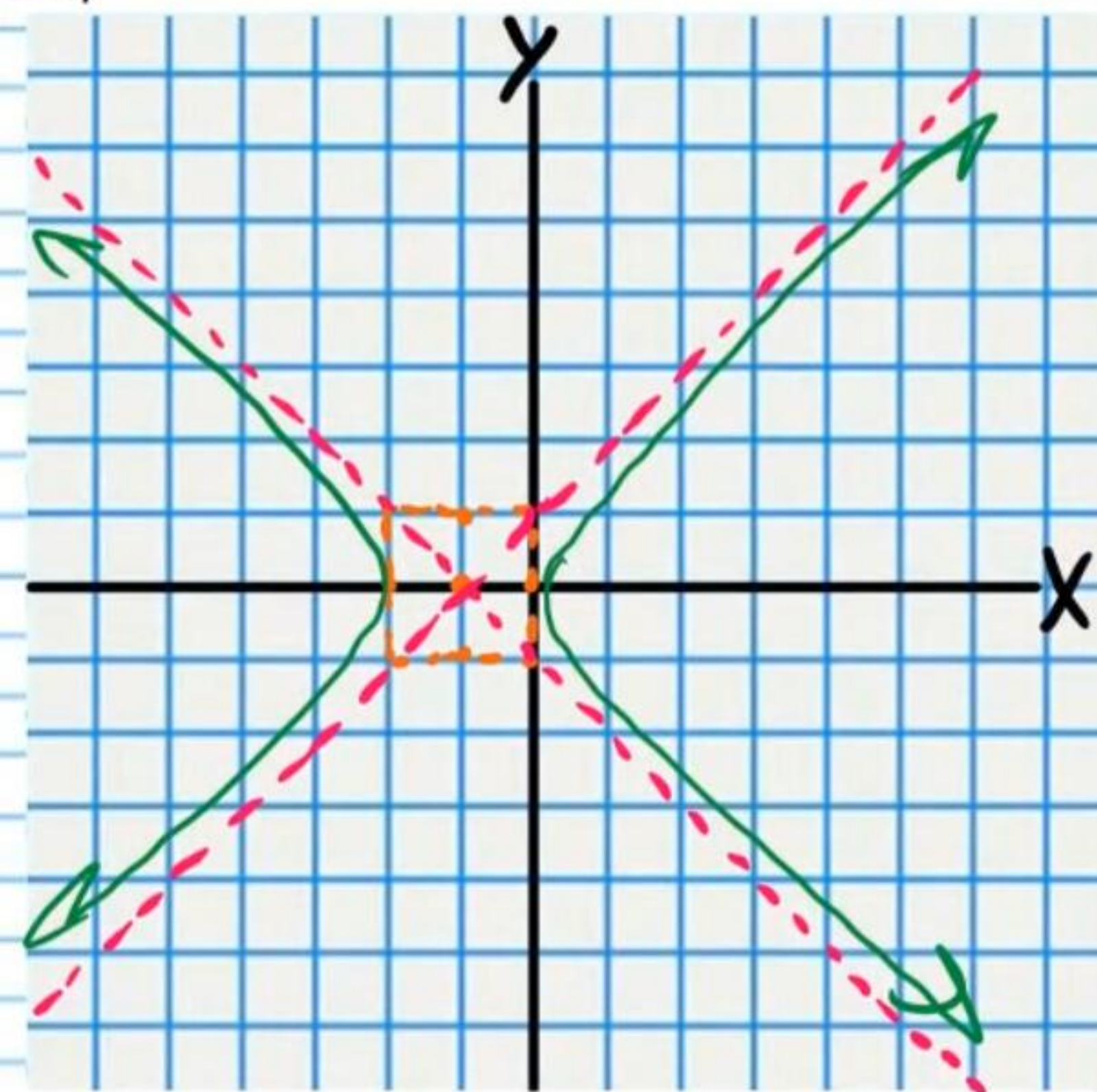
$$x^2 + 2x + \left(\frac{2}{2}\right)^2 - y^2 = 0 + \left(\frac{2}{2}\right)^2$$

$$x^2 + 2x + 1 - y^2 = 0 + 1$$

$$(x+1)(x+1) - y^2 = 1$$

$$(x+1)^2 - y^2 = 1$$

$$\frac{(x+1)^2}{1^2} - \frac{(y-0)^2}{1^2} = 1$$



$$\textcircled{10} x^2 + 4y^2 - 32y + 60 = 0$$

$$x^2 + 4y^2 - 32y = -60$$

$$x^2 + 4(y^2 - 8y) = -60$$

$$x^2 + 4(y^2 - 8y + (-\frac{8}{2})^2) = -60 + 4(-\frac{8}{2})^2$$

$$x^2 + 4(y^2 - 8y + 16) = -60 + 4(16)$$

$$x^2 + 4(y-4)(y-4) = 4$$

$$x^2 + 4(y-4)^2 = 4$$

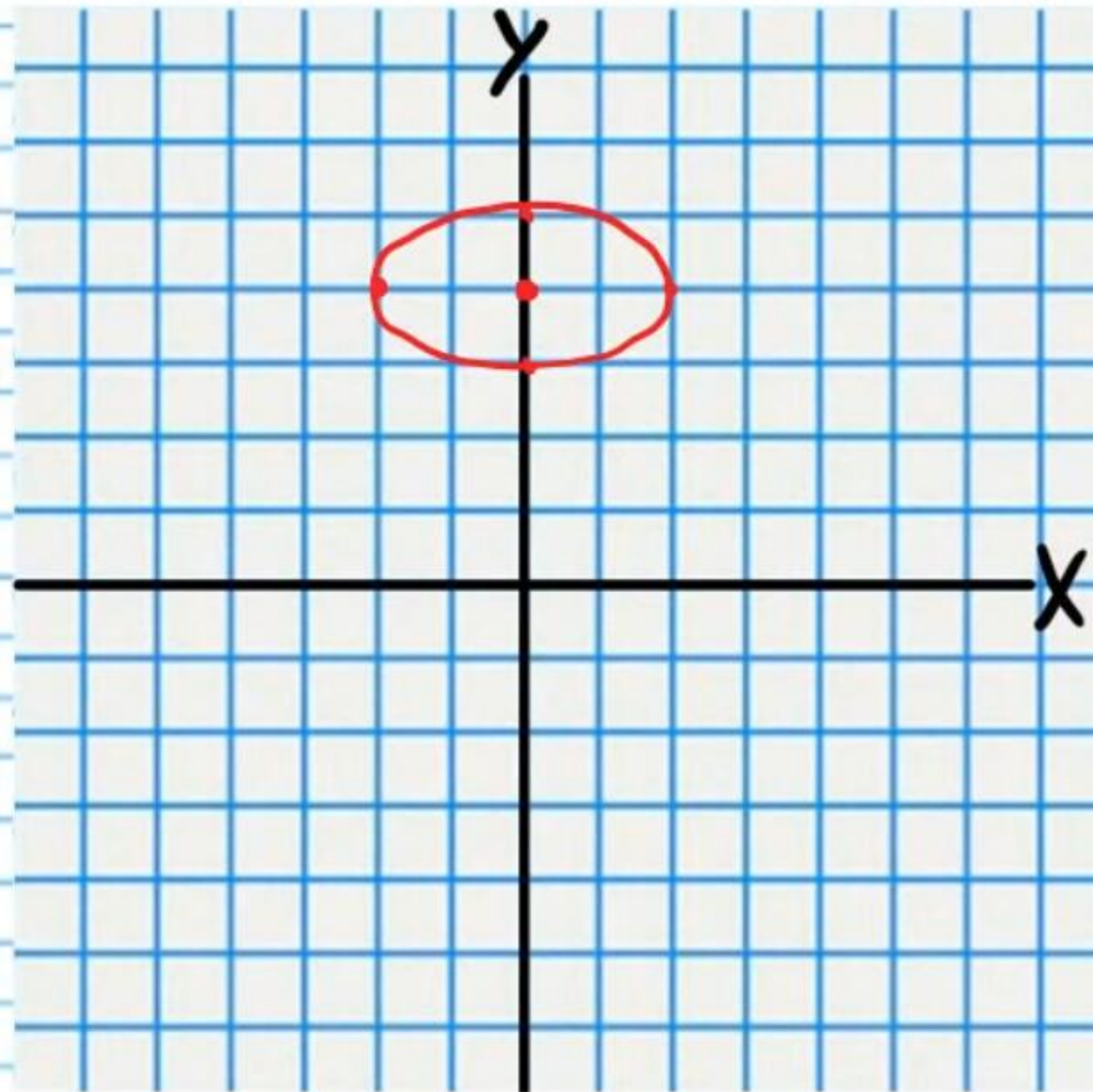
$$\frac{x^2 + 4(y-4)^2}{4} = \frac{4}{4}$$

$$\frac{x^2}{4} + \frac{4(y-4)^2}{4} = 1$$

$$\frac{x^2}{4} + (y-4)^2 = 1$$

$$\frac{(x-0)^2}{4} + (y-4)^2 = 1$$

$$\frac{(x-0)^2}{2^2} + \frac{(y-4)^2}{1^2} = 1$$



$$\textcircled{11} 25y^2 - 16x^2 - 100y - 300 = 0$$

$$25y^2 - 100y - 16x^2 = 300$$

$$25(y^2 - 4y) - 16x^2 = 300$$

$$25(y^2 - 4y + (-\frac{4}{2})^2) - 16x^2 = 300 + 25(-\frac{4}{2})^2$$

$$25(y^2 - 4y + 4) - 16x^2 = 300 + 25(4)$$

$$25(y-2)(y-2) - 16x^2 = 400$$

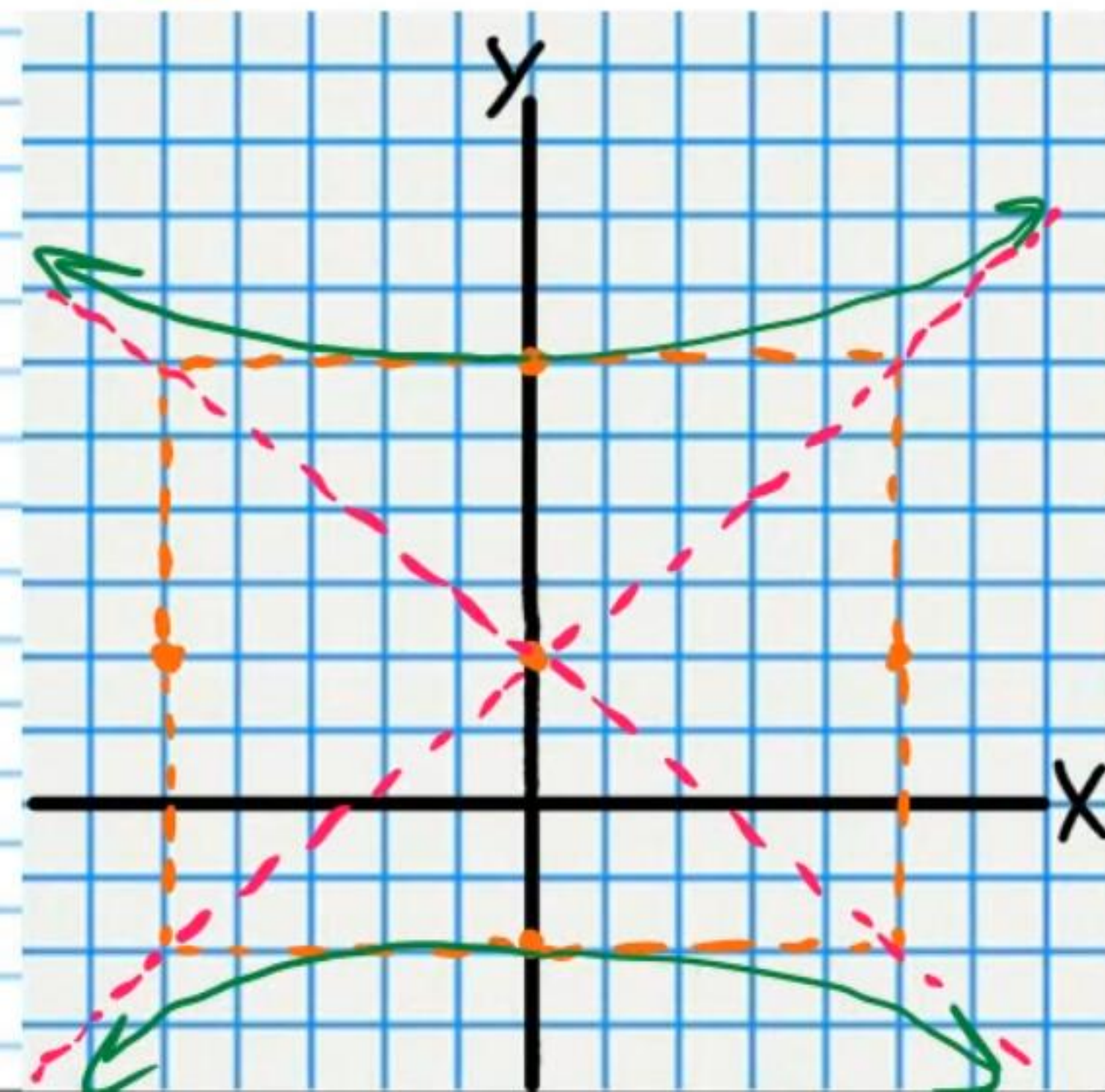
$$25(y-2)^2 - 16x^2 = 400$$

$$\frac{25(y-2)^2}{400} - \frac{16x^2}{400} = \frac{400}{400}$$

$$\frac{25(y-2)^2}{400} - \frac{16x^2}{400} = 1$$

$$\frac{(y-2)^2}{16} - \frac{x^2}{25} = 1$$

$$\frac{(y-2)^2}{4^2} - \frac{(x-0)^2}{5^2} = 1$$



$$\textcircled{12} \quad 4y^2 + 9x^2 + 72x + 108 = 0$$

$$4y^2 + 9(x^2 + 8x) = -108$$

$$4y^2 + 9(x^2 + 8x + \left(\frac{8}{2}\right)^2) = -108 + 9\left(\frac{8}{2}\right)^2$$

$$4y^2 + 9(x^2 + 8x + 16) = -108 + 9(16)$$

$$4y^2 + 9(x+4)(x+4) = 36$$

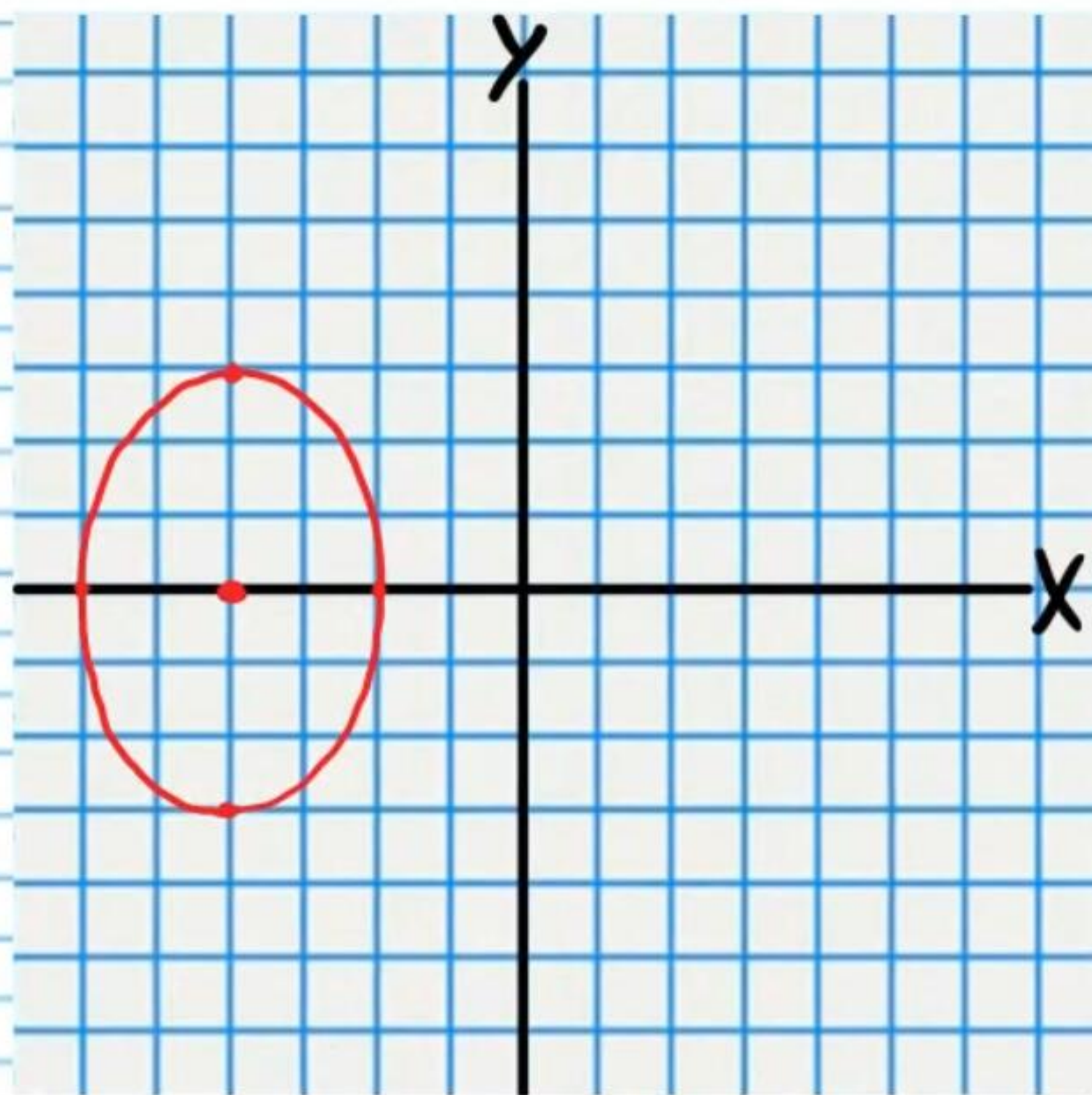
$$4y^2 + 9(x+4)^2 = 36$$

$$\frac{4y^2 + 9(x+4)^2}{36} = \frac{36}{36}$$

$$\frac{4y^2}{36} + \frac{9(x+4)^2}{36} = 1$$

$$\frac{y^2}{9} + \frac{(x+4)^2}{4} = 1$$

$$\frac{(y-0)^2}{3^2} + \frac{(x+4)^2}{2^2} = 1$$



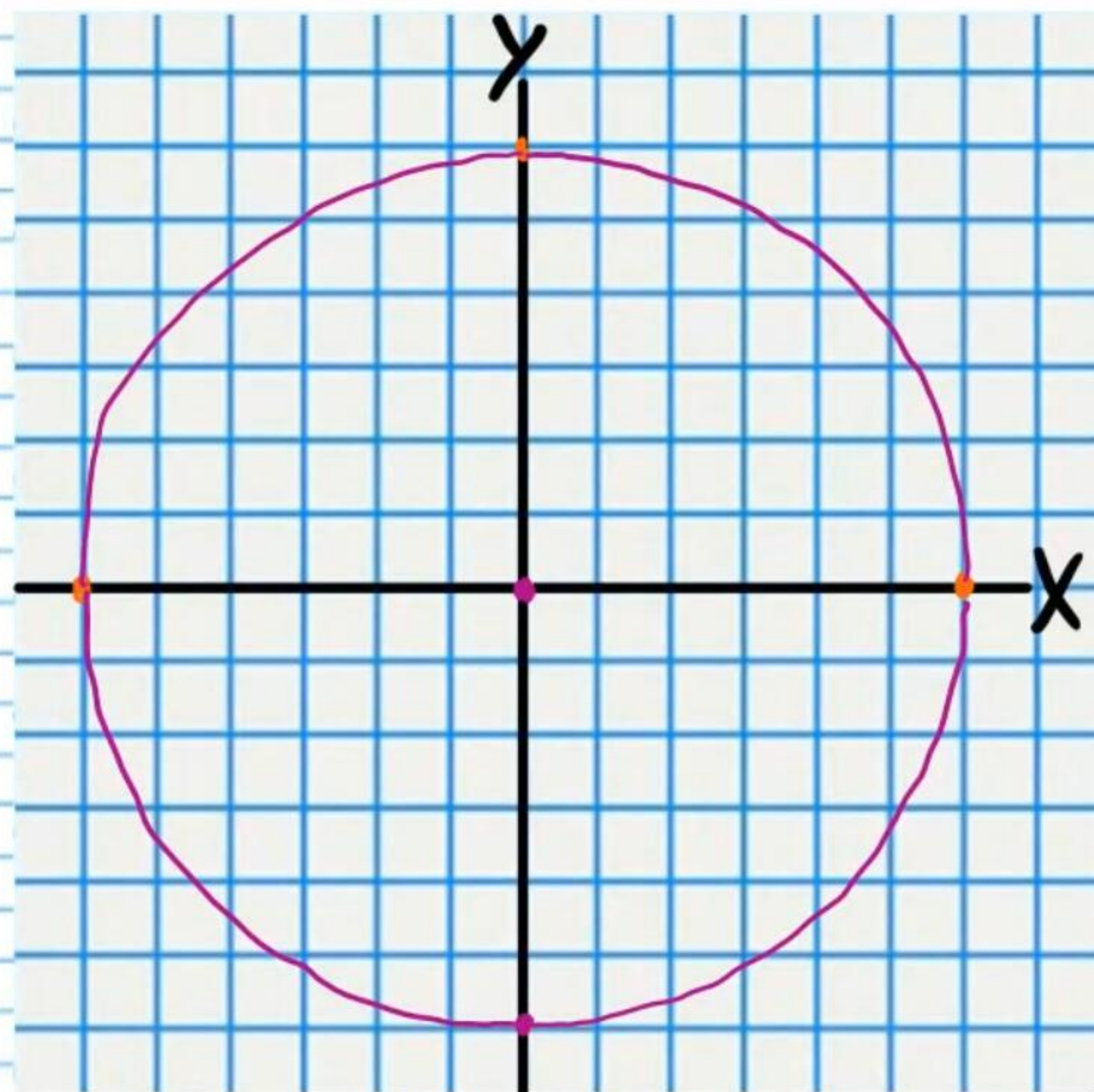
$$\textcircled{13} \quad \frac{x^2}{36} + \frac{y^2}{36} = 1$$

$$36\left(\frac{x^2}{36} + \frac{y^2}{36}\right) = 36(1)$$

$$\frac{36x^2}{36} + \frac{36y^2}{36} = 36$$

$$x^2 + y^2 = 36$$

$$(x-0)^2 + (y-0)^2 = 6^2$$



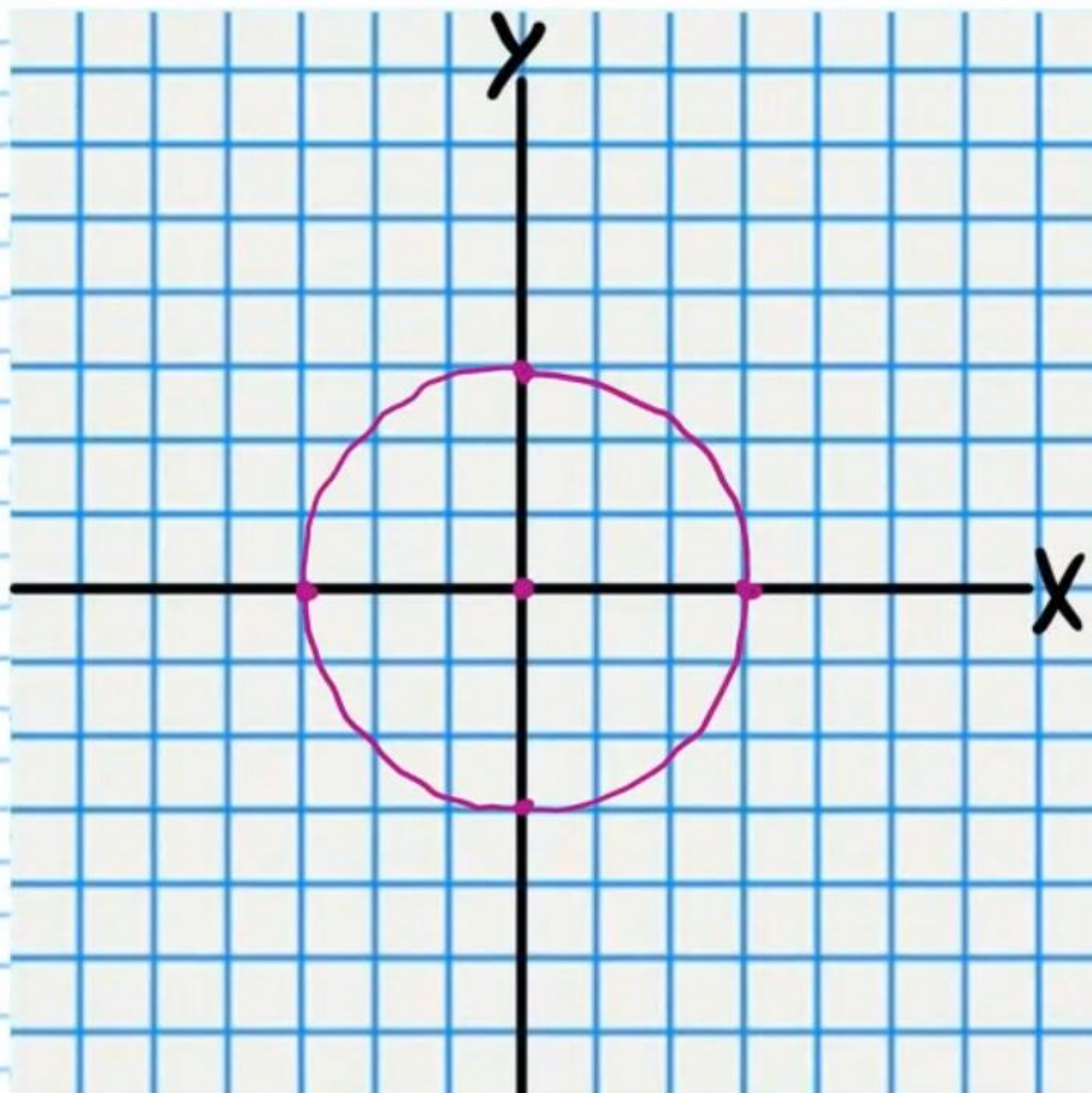
• (14) $3x^2 + 3y^2 = 27$

$$\underline{3x^2 + 3y^2 = 27}$$

$$\frac{3x^2}{3} + \frac{3y^2}{3} = \frac{27}{3}$$

$$x^2 + y^2 = 9$$

$$(x-0)^2 + (y-0)^2 = 3^2$$



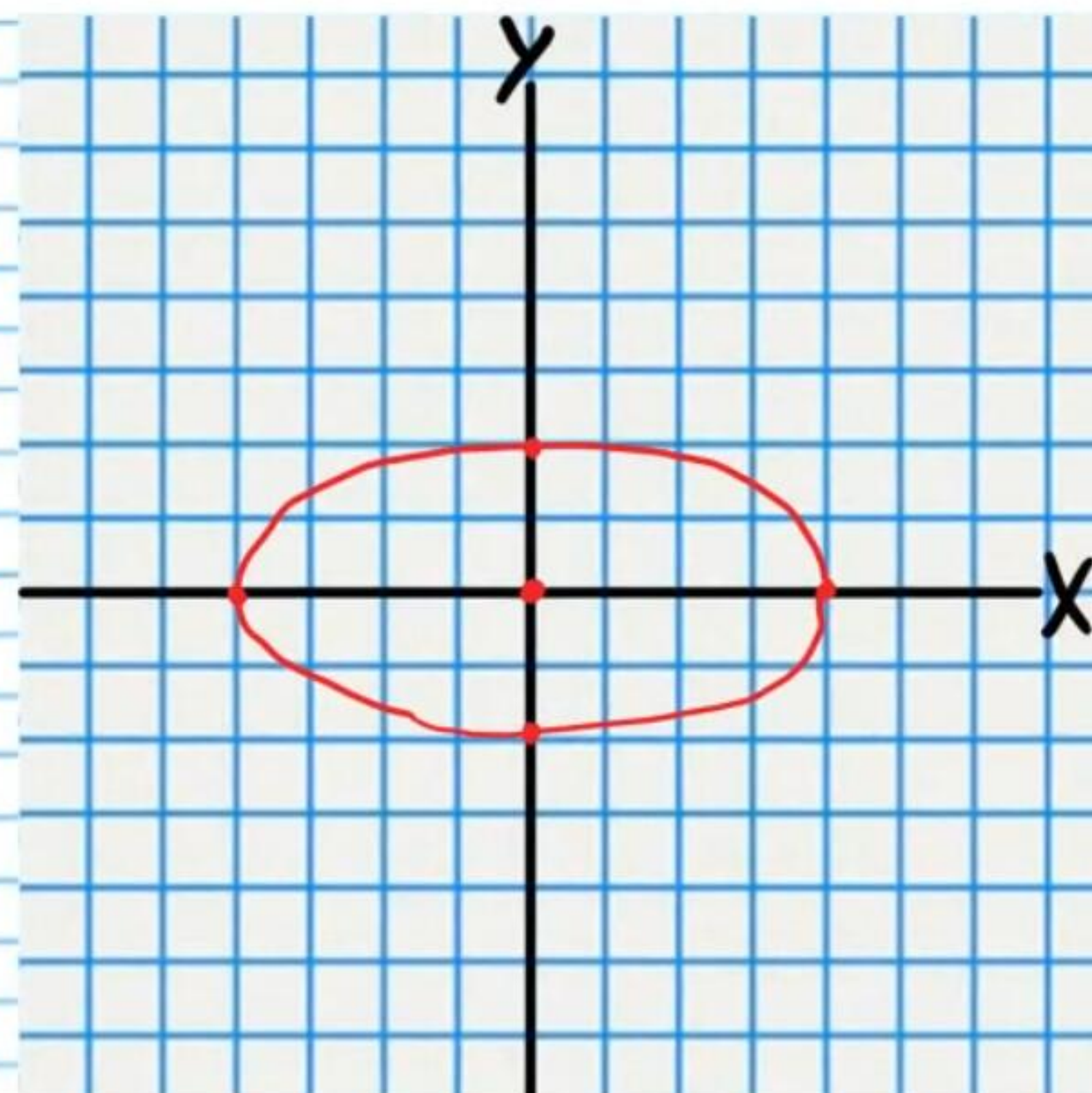
• (15) $4x^2 + 16y^2 = 64$

$$\underline{4x^2 + 16y^2 = 64}$$

$$\frac{4x^2}{64} + \frac{16y^2}{64} = \frac{64}{64}$$

$$\frac{x^2}{16} + \frac{y^2}{4} = 1$$

$$\frac{(x-0)^2}{4^2} + \frac{(y-0)^2}{2^2} = 1$$



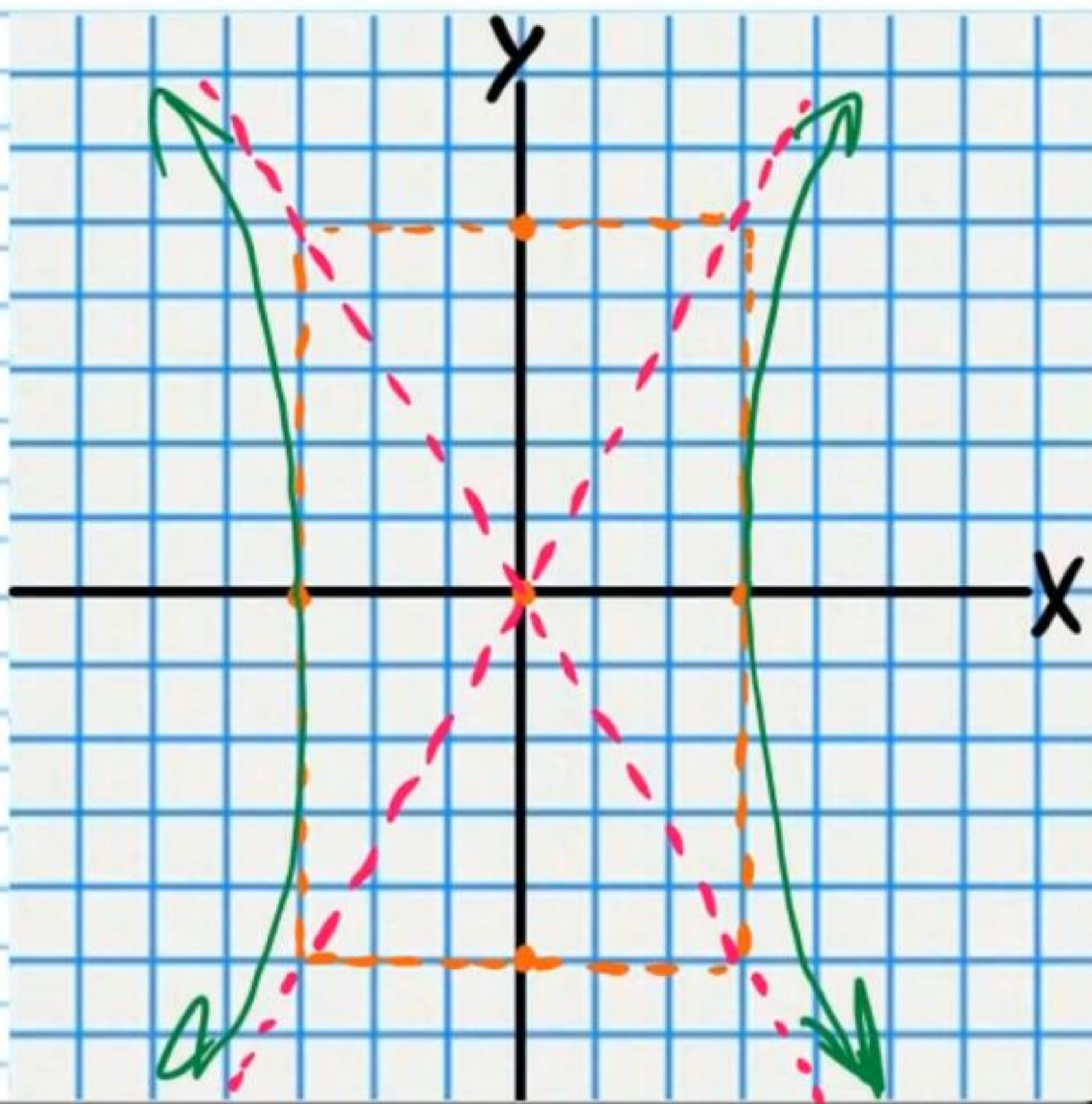
• ⑩ $25x^2 - 9y^2 = 225$

$$\frac{25x^2 - 9y^2}{225} = \frac{225}{225}$$

$$\frac{25x^2}{225} - \frac{9y^2}{225} = 1$$

$$\frac{x^2}{9} - \frac{y^2}{25} = 1$$

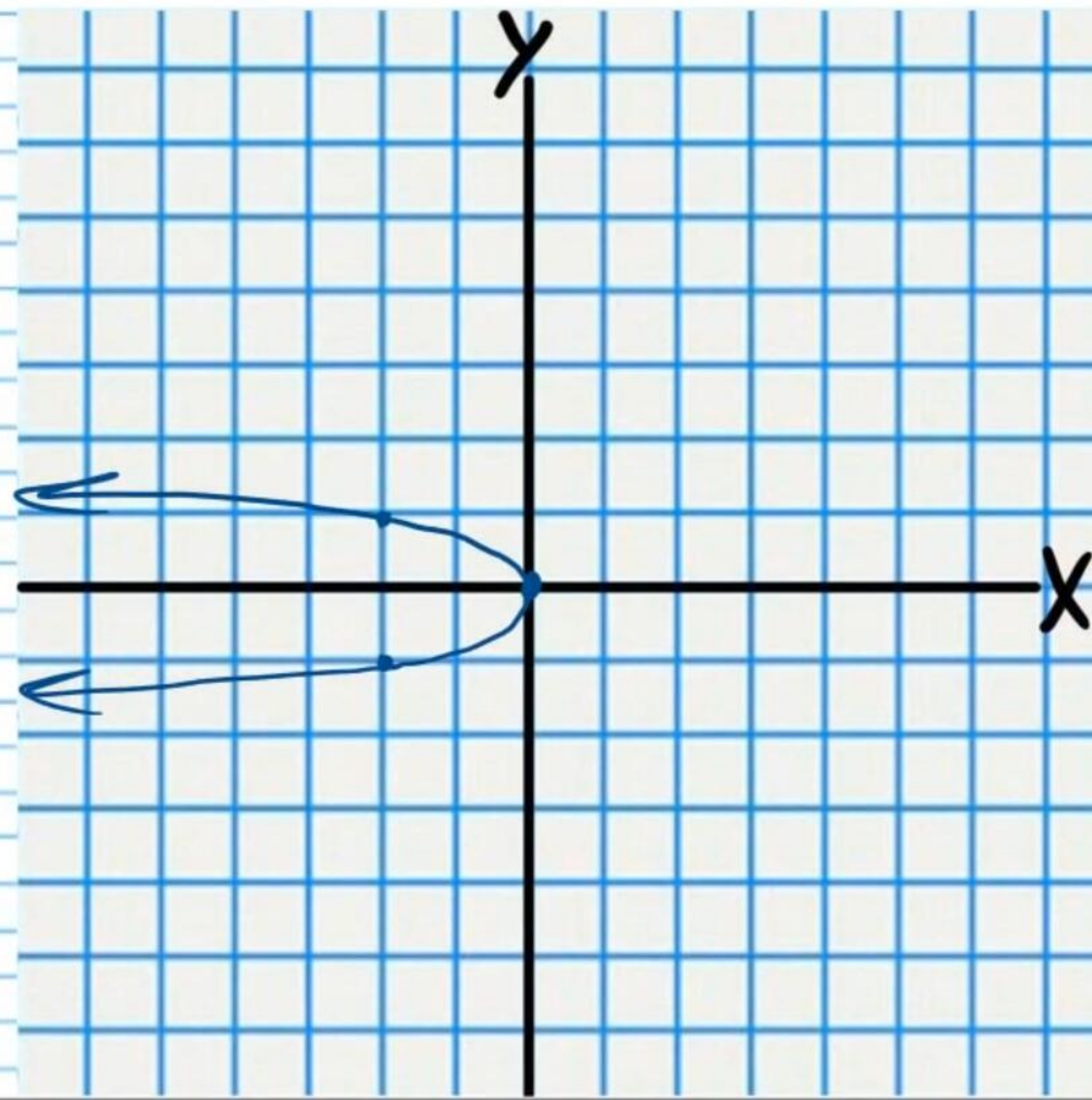
$$\frac{(x-0)^2}{3^2} - \frac{(y-0)^2}{5^2} = 1$$



• ⑪ $\frac{x}{-2} = y^2$

$$x = -2y^2$$

$$x - 0 = -2(y - 0)^2$$



$$\textcircled{18} -x^2 - 8x - 16 = y$$

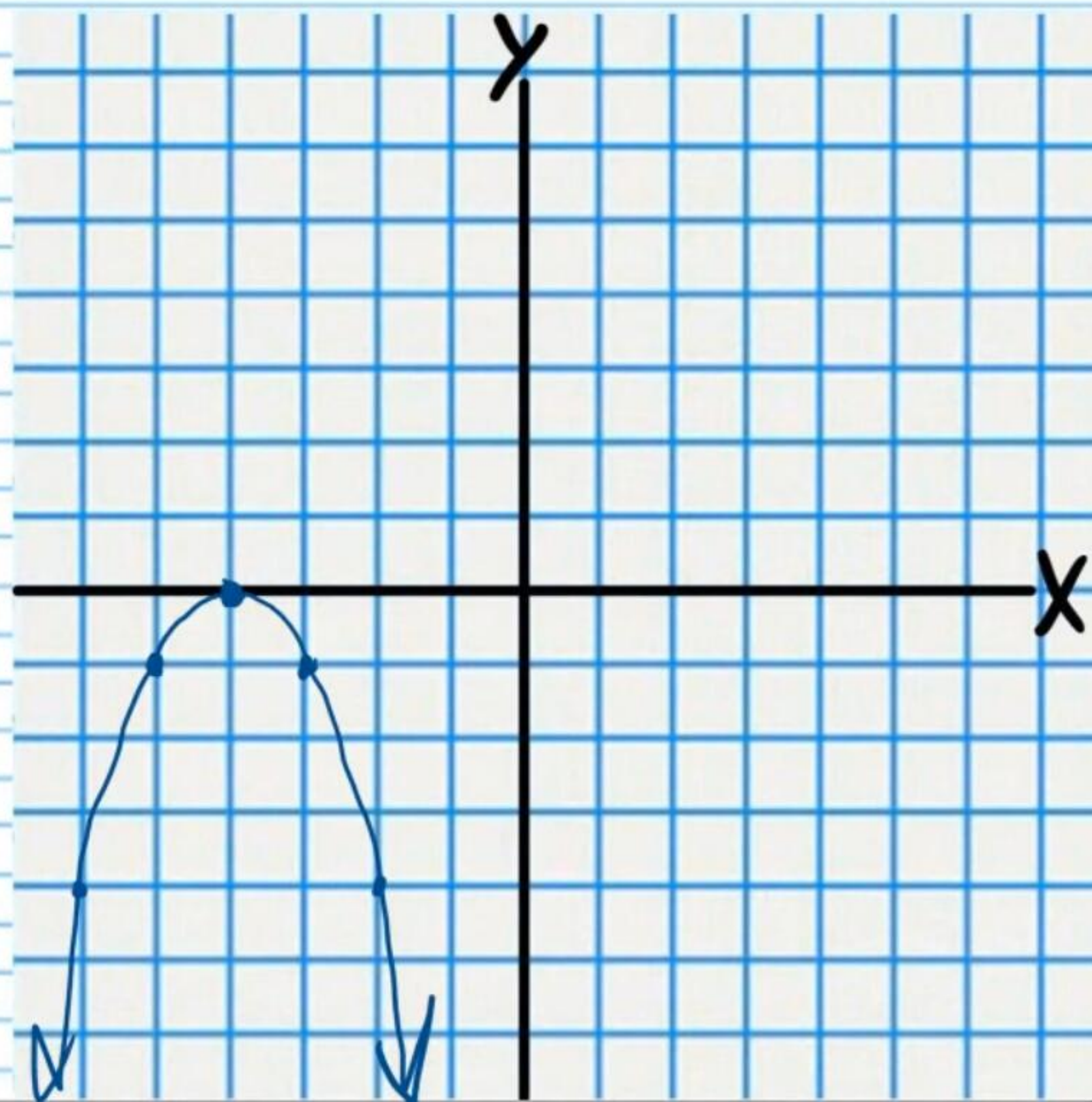
$$x^2 + 8x + 16 = -y$$

$$(x+4)(x+4) = -y$$

$$(x+4)^2 = -y$$

$$-(x+4)^2 = y$$

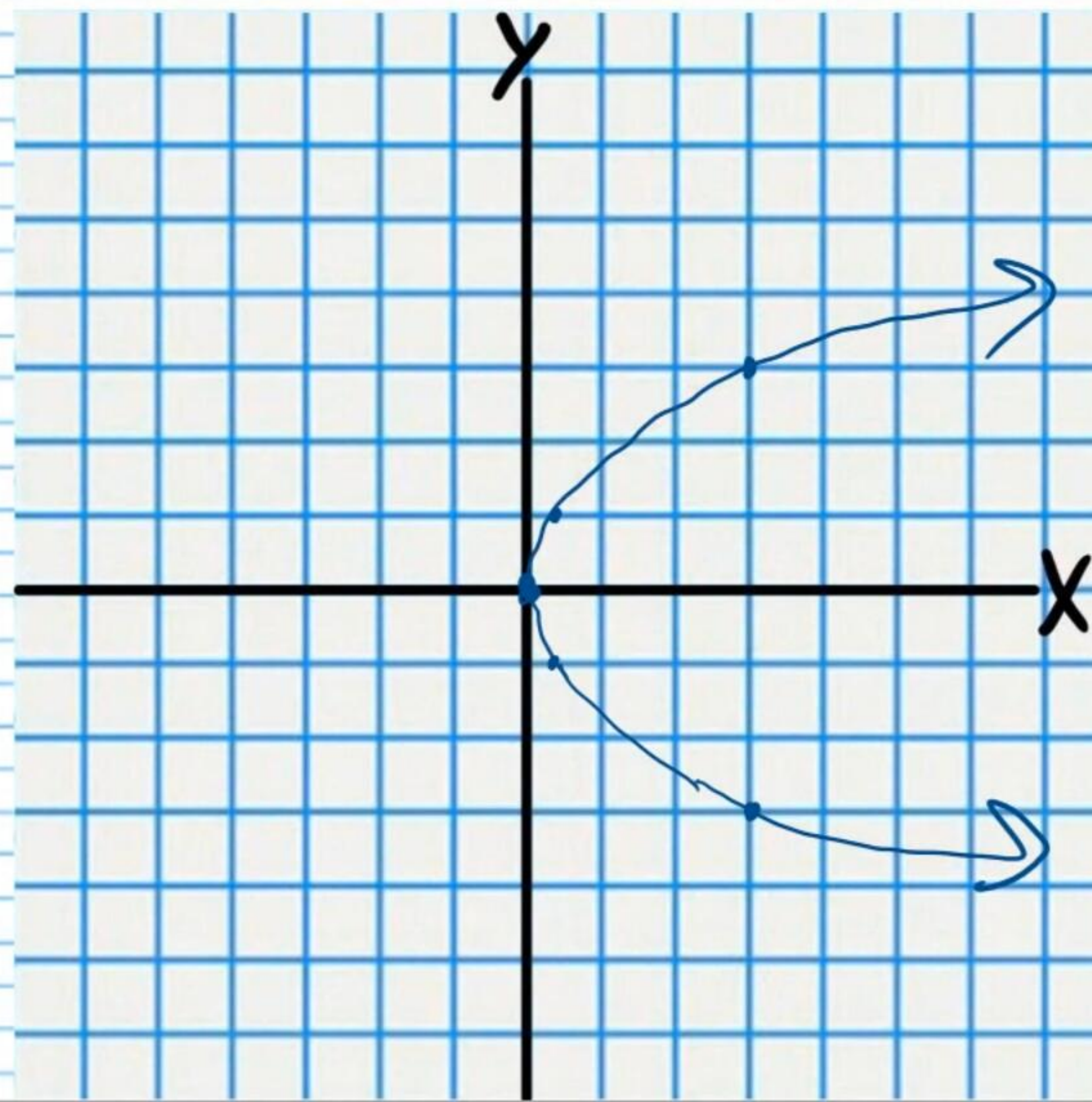
$$-(x+4)^2 = y - 0$$



$$\bullet \textcircled{19} 3x = y^2$$

$$x = \frac{1}{3}y^2$$

$$x - 0 = \frac{1}{3}(y - 0)^2$$



②① $\frac{y+1}{4} = x^2$

$$y+1 = 4x^2$$

$$y+1 = 4(x-0)^2$$

